

ENGM3000 Mechatronics Foundations Design Project

Aim

In this project, your aim is to simulate and analyse Watt's six-bar parallel motion linkage in Simulink and Simscape. The design should include a DC motor controlled by PWM signals and a H-bridge that is capable of switching direction and speed/torque control. The DC motor is connected to a gear train of certain dimensions to provide enough torque to the Watt linkage. At the end (output) of the linkage, a suspension acting like a spring-damper system is attached to measure the output force. Your task is as follows:

- Construct Watt's six-bar parallel motion linkage in Simscape
- Incorporate a DC motor circuit controlled by PWM signals and H-Bridges into the linkage
- Design a gear train which generates a specified amount of torque at the driver joint, denoted Fixed Joint 1 in Figure 1A.
- Analyse the amount of force at the linkage output given differencing cases of spring stiffness and damper coefficients of the suspension system

Specification requirements for the above are detailed in the following pages along with an appendix to assist with construction of the relevant Simscape subsystems.

Report requirements

This is an **individual project worth 30%**. You are required to submit a maximum 10-page report excluding appendices.

Your report should include the following sections:

- Introduction: state the aim and objectives of the project, briefly summarising your findings.
- DC motor circuit: explain the use of PWM signals and H-Bridge to control DC motor speed and direction. Include any relevant schematic, model and figures.
- Gear train: show calculations for your gear train design, from the input torque to the desired output torque. Include any relevant schematic, model and figures.
- Watt's linkage: explain the role of Watt's parallel motion linkage and the mechanism behind vertical motion of the linkage output. Discuss the maximum vertical distance the linkage output can achieve. Include a figure showing your linkage in Simulink.
- Suspension: show the results of your case study. Discuss the role of spring stiffness and damping coefficient on the force profile of the linkage output. Include any relevant schematic, model and figures supporting your results.
- Reflection: reflect on the use of MATLAB/Simulink and Simscape as tools to simulate electrical and mechanical components, discussing both advantages and shortcomings.
- Conclusion: summarise the function of your system and results

Submission is via the myLearning (Canvas) site for Mechatronics Foundations under the Design Project assessment. Please upload a **pdf file** of your report and your **Simscape/Matlab files** which will be run to ensure they work.

The due date is **Friday 29 May, 23.59 PM**

Watt's parallel motion linkage

Devised by Scottish engineer James Watt, the Watt parallel motion linkage is a six-bar mechanism that transforms rotational motion to near-linear motion. Originally invented in 1784 as part of his steam engine that contributed significantly to the industrial revolution, this linkage now is a constituent mechanism in automotive suspension systems. Specifically, it prevents the axle from moving side-to-side relative to the vehicle chassis during cornering. With the correct link lengths and position of the fixed joint, the output of the linkage can achieve linear up and down motion. One example of Watt's parallel linkage is shown in Figure 1A schematic, the corresponding Simulink model is shown in Figure 1B. Note that Joint 3-4 and Joint 3-6 are separated joints but located at the same point (link 3 is connected to both link 4 & 6).

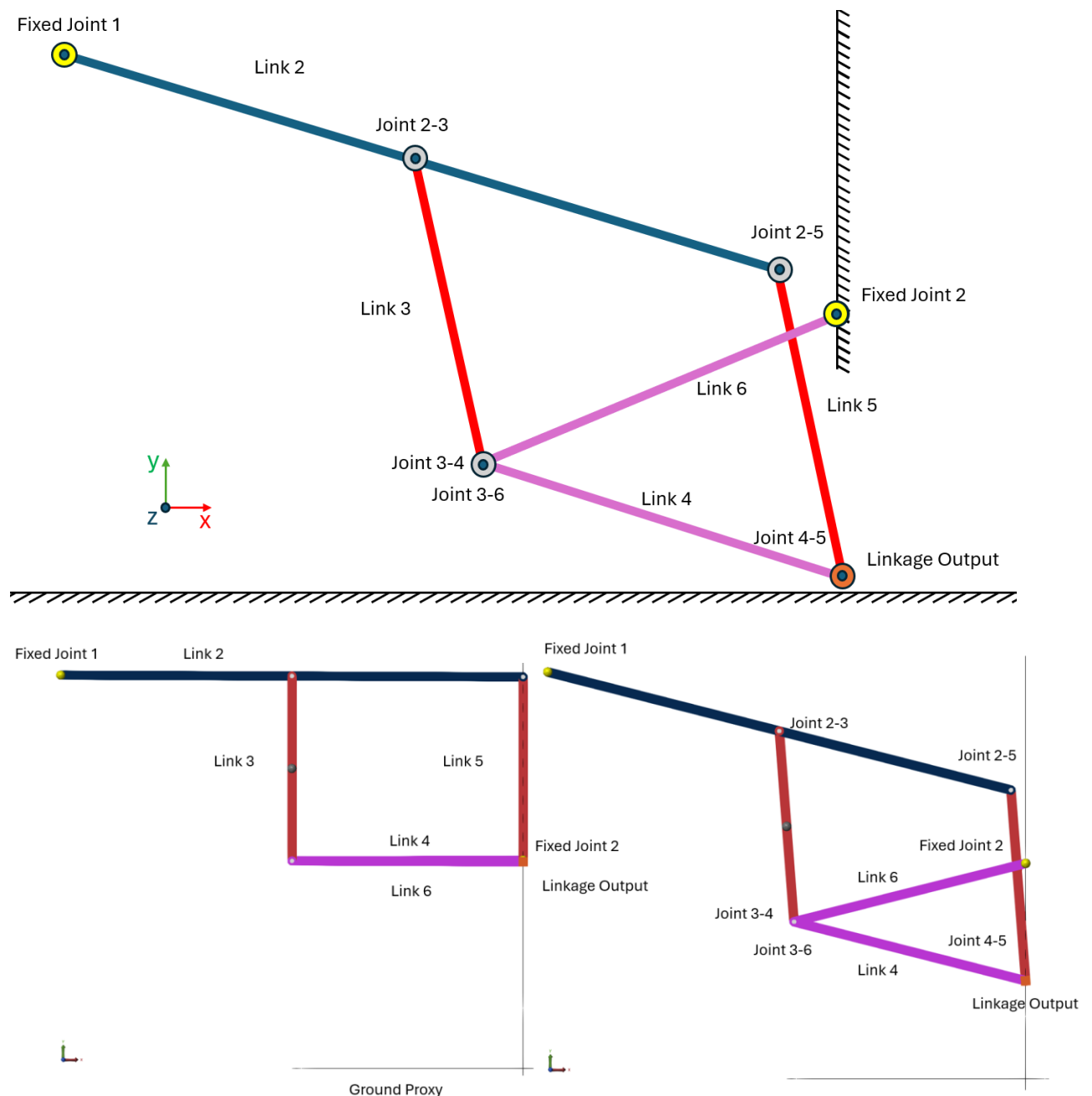


Figure 1A: Watt's parallel linkage schematic.

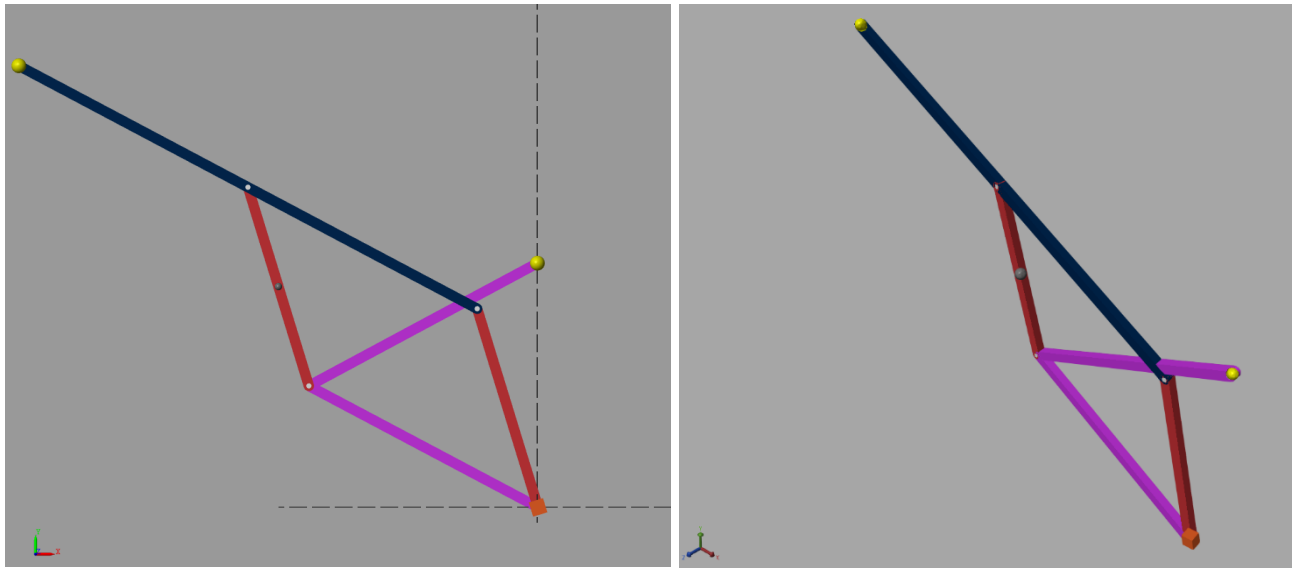


Figure 1B: Watt's parallel linkage Simulink simulation.

The linkage has 6 links and 7 joints. The two fixed joints are revolute joints that cannot be translated in Cartesian space (X-Y-Z frame). Link 2 serves as the motion driver (driver joint) of the linkage, i.e. fixed joint 1. Consider the total distance between the two fixed joints (fixed joint 1 and 2) as link 1, Table 1 denotes the link parameter for each link:

Table 1: Link parameters

Name	Colour	Link length	[Width (m), Height (m), Mass (kg)]
Link 2 (from Fixed Joint 1 to Joint 2-5)	Blue	$a + a = 1 \text{ m}$	[0.02, 0.02, 2]
Link 3 (from Joint 2-3 to Joint 3-4) & 5 (from Joint 2-5 to Joint 4-5)	Red	$b = 0.4 \text{ m}$	[0.02, 0.02, 1]
Link 4 (from Joint 3-4 to Joint 4-5) & 6 (from Joint 3-6 to Fixed Joint 2)	Purple	$a = 0.5 \text{ m}$	[0.02, 0.02, 1.2]
Fixed joints 1 & 2 vertical distance		$\approx b$	
Fixed joints 1 & 2 horizontal distance		$a + a$	
Fixed joints 1 vertical distance to ground		0.85 m	

DC motor, PWM and H-Bridges:

In the lecture content, you were introduced to how PWM signals can be used alongside H-Bridges to control speed and direction of a DC motor. For the electrical side of the model, you are required to create a circuit that can control the DC motor, such that the motor can operate at a desired speed and operate in clockwise and anticlockwise direction. The motor specification is provided in Table 2.

Table 2: Motor Specification

Electrical	
No-load speed	2000 rpm
Rated speed (at rated load)	1000 rpm

Rated load (mechanical power)	750 W
Rated DC supply voltage	48 V
Mechanical	
Rotor inertia	0.0002 kg*m ²
Rotor damping	0.8e-5 N*m*s/rad

Suspension – Measuring force at the linkage output

The role of a suspension is to absorb force between two bodies. Instead of having the linkage output hitting the ground, which would result in a large impact force on the linkage, a suspension is placed between the linkage output and ground. The suspension acts as a mass-spring-damper system. When forces are applied, the suspension stretches or compresses. Depending on the spring distance travelled and the spring stiffness and damping coefficient, the applied force can be calculated.

Design Task

Design an electrical circuit in Simscape that utilises PWM signals and H-Bridges to control the given DC Motor. Monitor and plot the **motor speed (no-load speed)** at a PWM **duty cycle <85%** while **switching direction every 5 seconds**. Also plot the **torque profile generated by this PWM signal**. Note the amount of torque output by the motor.

You are to design a gear train to receive the torque output from your DC motor at a **duty cycle <85%**. The torque profile **switches direction every 5 seconds**. The gear train output shaft is connected to link 2 of the Watt parallel linkage, providing torque to the driver joint (i.e. Fixed Joint 1). The torque received at Fixed Joint 1 of link 2 must be around **100 Nm**. **The gear radii must be within the range of $4\text{ cm} \leq r_i \leq 10\text{ cm}$.**

Your gear train must **physically** connect to **Fixed Joint 1 of link 2 of the linkage** to provide torque. Include in your report the calculation of your gear ratio, and your working showing how to get to 100 Nm, from the initial torque from the DC motor. The input torque (measured at the gear train driver joint) and the output torque (measured at **Fixed Joint 1 of link 2** connected to the gear train output shaft) must be plot on the same figure, clearly showing the step-up in torque that matches your gear ratio.

Case study

In this case study, you are given different combinations of suspension parameters to simulate. Modify your model and analyse the force profile at the linkage output. Use the output of the **Spring and Damper Force block** and **Spatial Contact Force block** as tools for your analysis. You can enable more output as you see fit, such as ground force, enabled from the Spatial Contact Force block (when the spring is too weak to resist the output force and the linkage hit the ground). Also, it's required to measure to **input torque at Fixed Joint 1 of link 2**.

Table 3: Different suspension parameters and torque input for case study

Case	Input Torque Magnitude (Nm)	Input Torque Frequency (Hz)	Spring Stiffness (Nm)	Damper Coefficient (N/(m/s))	Spring natural length (m)
1	100	0.2	1000	0	0.3
			750		
			500		
2	100	0.2	1000	20	0.3
				50	
				100	
3	200	0.5	?	?	0.3

Run your model for 20s and export the **torque input, force profile and distance travelled** to MATLAB and plot them together for each case. Plot the data in a clear and concise manner. For each case, provide a discussion on the behaviour of the force. Your discussion should include:

- In case 1, was the force “bouncy” or damped? Was the spring stiffness enough to stop the output from hitting the ground? Remember the function of the suspension is to absorb incoming forces and prevent bodies from hitting each other. Would it be good for a suspension to behave like these cases?
- In case 2, what do you notice about the force profile when the damping coefficient increases? Can we keep increasing the damping? What is the most suitable combination of spring stiffness and damping ratio, given that we want the suspension to absorb force as quick and steady as possible? Use this combination for case 3 initially.
- For case 3, how did you increase the torque magnitude and frequency? Provide the force profile and distance travelled when applying the stiffness and damping combination in case 2? Was it enough to absorb all force? Form your own analysis of this case and include necessary data and figures to choose the most appropriate stiffness and damping coefficients.

Appendix

General tip:

1. Use the previous built gear train and linkage (crank, coupler and rocker) and modify them with the Watt's linkage dimension. Ensure you get the length and mass right.
2. Build each model separately and test for its function before combining them together. You should always have a copy of each model just in case your combined model stops working.
3. Always measure and plot input/output of the models and you may include these plots in your report to support your understanding.

Watt's linkage

- Construct the parallel linkage in Simscape according to the given parameters. Apply a torque at the driver joint (Link 2's Fixed Joint 1). All joints in the linkage must have a damping coefficient of **0.02 N*m/(deg/s)**. Place a block called **Point** from the Simscape Multibody library where joint 4 and 5 link together (Linkage Output in Figure 1) to monitor the vertical motion of the linkage output. This block acts as a proxy which allows you to measure the distance from ground and force at the linkage output.
- As the driver joint moves, the output point moves linearly and vertically from the ground. Note down the amount of torque required to lift the whole linkage up. This is the least amount of torque that is required to move the linkage. **Important:** you can set the initial joint angle of Fixed Joint 1 of link 2 by Specify Position Target under State Target, set this to -10 degrees for ease of simulation later.
- To setup an environment to analyse this linkage output, we are going to create a ground proxy for the linkage output to interact with. You will need an **Infinite Plane block** and a **Spatial Contact Force block** to create a ground proxy. The Infinite Plane creates a visual for your ground while the Spatial Contact Force simulates forces acting between two rigid bodies. The distance from the top of the linkage (Fixed joint 1) and the Infinite Plane should be **0.85m**. In the Spatial Contact Force block, set the damping to **1e3 N(m/s)** and **Stiffness to 1e6 N/m**. Under sensing, explore the different parameters you can output to Simulink scope, noticeably **Separation Distance** and **Normal Force magnitude**. Important note: once created, ensure the z-axis of the infinite plane points upwards by setting the Rotation in a Rigid Transform block. You might also want to offset this plane from the World Frame to be directly below the output point.

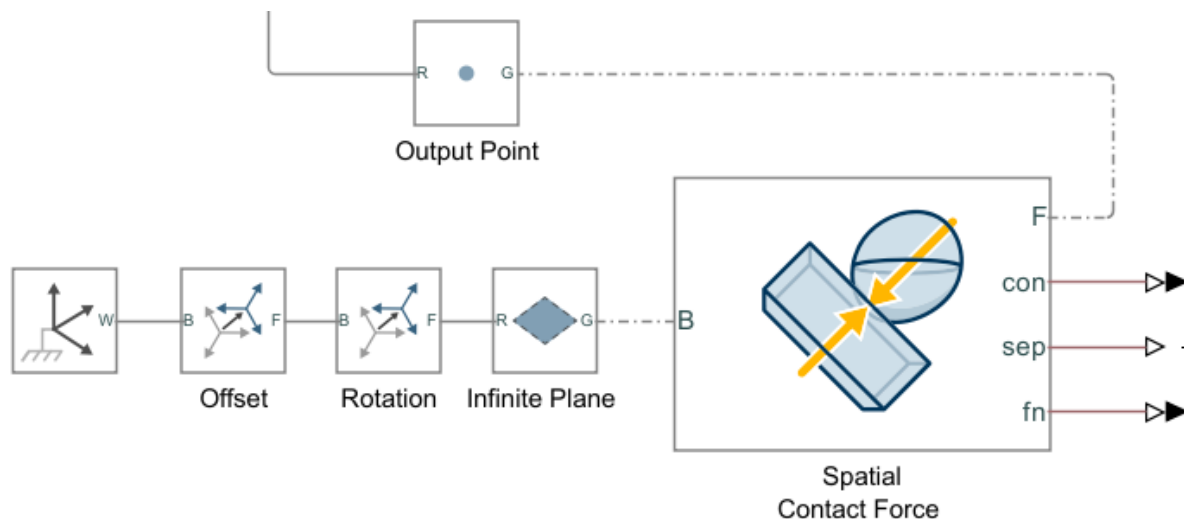


Figure 2: Ground and Contact Force setup

- If set up correctly; remove the input torque and the output of the linkage will rest on the ground as it falls, as shown in Figure 1. You might also notice that the output only travels vertically linearly for a certain distance as it is dependent on the link parameters and locations of the fixed joints. Experiment with the location of these two fixed joints to see the effect on the linear distance of the output.

DC motor, PWM generator and H-bridge

As shown in Figure 2, Simscape Electrical provides convenient Controlled PWM Voltage and H-Bridge blocks to use, without building the entire electrical circuit from scratch.

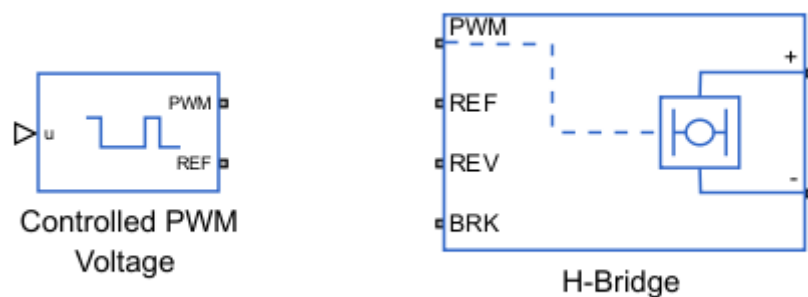


Figure 3: Controlled PWM Voltage block and H-Bridge block

For this project, utilise **Averaged** for the **Simulation mode** in the PWM block and specify use **PS input** for the **Modelling option**. The input of the PWM block is the desired duty cycle of the output signal. Changing this duty cycle varies the average output voltage of the generated PWM output. Set the PWM frequency to 4000 Hz. This frequency doesn't affect the averaged voltage, which is determined by the duty cycle, but it does affect how the motor's internal inductance reacts to the drop in current. In real life application, this frequency sits at 1kHz to 20 kHz, high enough for the inductance to smooth out the current, providing a smooth output torque. Connect the input to a

Constant block and specify a value between 0 and 1 to change the duty cycle. Monitor the output voltage of the PWM as the duty cycle changes.

Similarly, for the H-Bridge, set the **Simulation mode** to **Averaged**. The H-Bridge block consists of four inputs, with those of interest being the PWM and REV inputs. PWM specifies the output voltage amplitude from the H-Bridge, depending on the PWM duty cycle set earlier. The REV input changes the output voltage direction when the voltage at REV port goes above or below a certain threshold, hint: use this port to switch the direction of the motor. Inside the H-Bridge block, you can set **reverse threshold** and **PWM signal amplitude** as desired. For this circuit to work properly, you may need **two PWM blocks**, one for PWM input of the H-Bridge and the other for the REV port. The **Signal generator** block from Simulink might be helpful for the PWM block that connects to the REV port. From the H-Bridge, you are required to create a square voltage profile at **$\pm 48V$ amplitude with 0.1 Hz frequency**.

The DC motor given in Table 2 is standard for almost all commercial motors. To simulate such specification, change the **Model parameterization** in the **DC Motor block** to **By rated load and speed** and enter the given parameters. Connect the PWMs, H-Bridge and DC Motor; use one of the two blocks below to measure the motor speed and torque. You don't need to add the Gear Box block and Rotational Friction block.

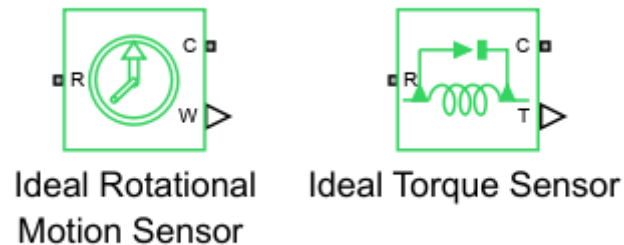


Figure 4: Motion sensor and Torque sensor

Note: only use one at a time, the Ideal Torque Sensor measures torque when the motor is stalled, which results in 0 rpm speed.

An example of H-Bridge voltage, motor speed and torque at **100% duty cycle at 0.1 Hz frequency** is shown below:

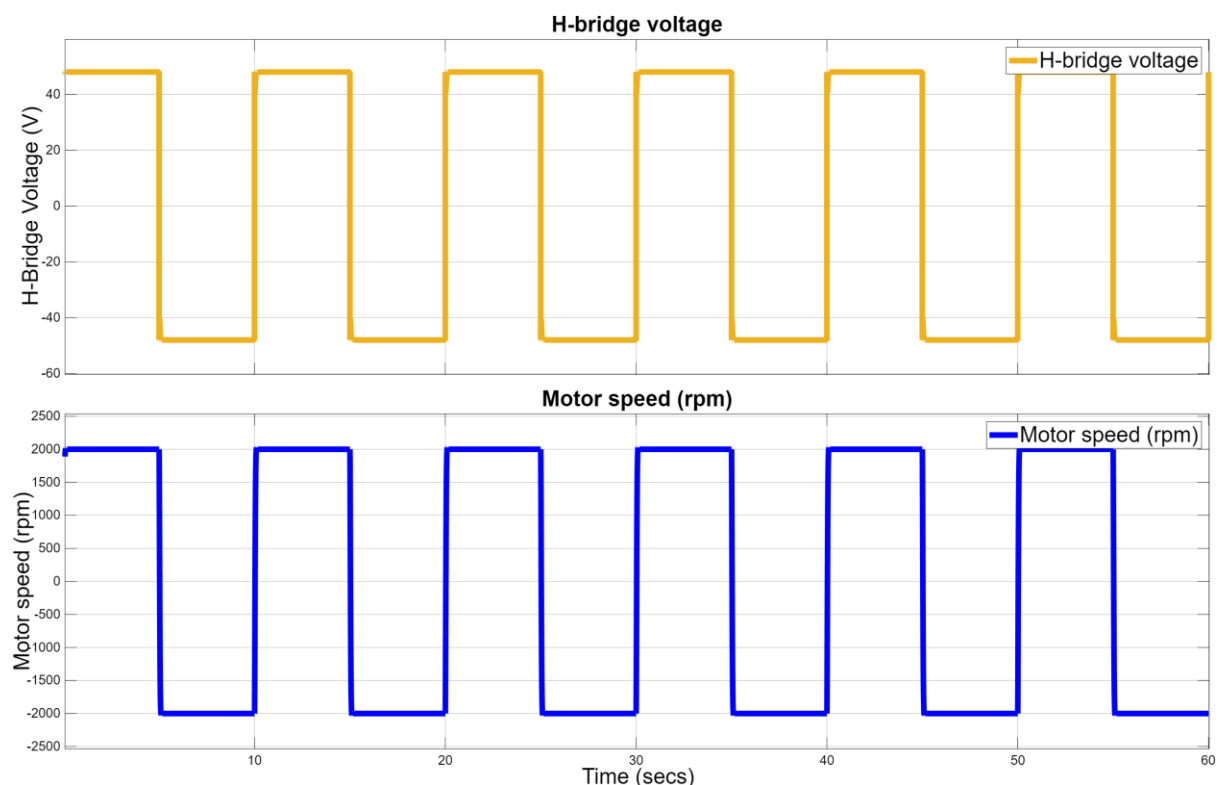


Figure 5: H-Bridge Voltage vs Motor speed

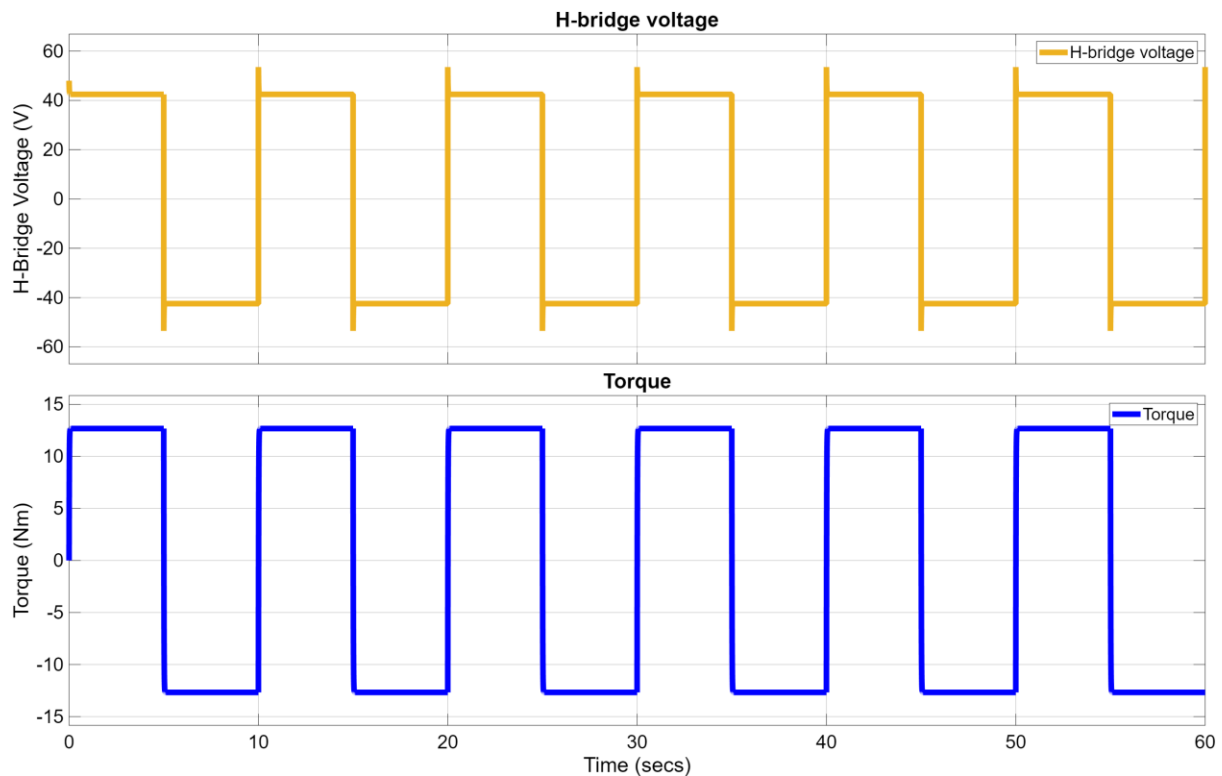


Figure 6:H-Bridge voltage vs Torque

Suspension

The **Spring-Damper Force block** can simulate the suspension behaviour **and** enables for the vertical force component at the linkage output to be measured. The block creates a spring-damper relationship between two bodies when connected. When one body provides force onto the other, you expect to see a spring-like force profile between them. Refer to Figure 7 for suggested placement. The **natural length and other parameters** of the spring-damper can be freely chosen; for testing you can set it at **0.3 m** and spring stiffness at **1000 Nm** while keeping the **damper at 0**. You can then enable force sensing within this block to measure force as the linkage output moves toward the ground proxy.

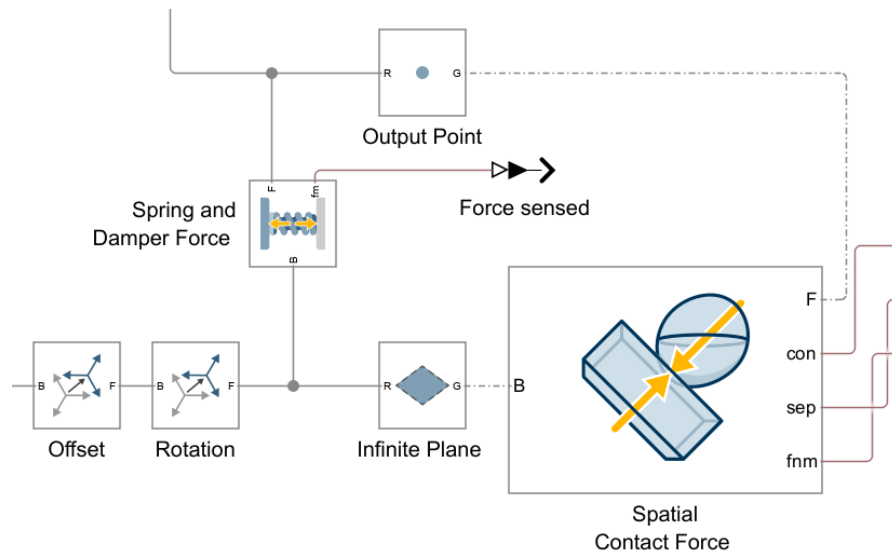


Figure 7: Spring-damper connection

Investigate the force behaviour of the system by changing the spring stiffness and spring's natural length. You will need to provide a short discussion on your understanding of these behaviours while generating force profile for each case in the report. You can use the **Separation Distance (sep)** output of the Spatial Contact Force block to measure the **distance the linkage output travelled (from ground)**.